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#pragma HLS interface <mode> port=<name> bundle=<string> \
register register_mode=<mode> depth=<int> offset=<string> \
clock=<string> name=<string> \
num_read_outstanding=<int> num_write_outstanding=<int> \
max_read_burst_length=<int> max_write_burst_length=<int>
```

Where:

- **<mode>** : Specifies the interface protocol mode for function arguments, global variables used by the function, or the block-level control protocols. For detailed descriptions of these different modes see "Interface Synthesis Reference" in the *Vivado Design Suite User Guide: High-Level Synthesis* (UG902 (<https://www.xilinx.com/cgi-bin/docs/rdoc?v=2019.1;d=ug902-vivado-high-level-synthesis.pdf>)). The mode can be specified as one of the following:
 - **ap_none** : No protocol. The interface is a data port.
 - **ap_stable** : No protocol. The interface is a data port. The HLS tool assumes the data port is always stable after reset, which allows internal optimizations to remove unnecessary registers.
 - **ap_vld** : Implements the data port with an associated `valid` port to indicate when the data is valid for reading or writing.
 - **ap_ack** : Implements the data port with an associated `acknowledge` port to acknowledge that the data was read or written.
 - **ap_hs** : Implements the data port with associated `valid` and `acknowledge` ports to provide a two-way handshake to indicate when the data is valid for reading and writing and to acknowledge that the data was read or written.
 - **ap_ovld** : Implements the output data port with an associated `valid` port to indicate when the data is valid for reading or writing.

IMPORTANT: The HLS tool implements the input argument or the input half of any read/write arguments with mode `ap_none`.

- **ap_fifo** : Implements the port with a standard FIFO interface using data input and output ports with associated active-Low FIFO `empty` and `full` ports.

Note: You can only use this interface on read arguments or write arguments. The `ap_fifo` mode does not support bidirectional read/write arguments.

- **ap_bus** : Implements pointer and pass-by-reference ports as a bus interface.
- **ap_memory** : Implements array arguments as a standard RAM interface. If you use the RTL design in the Vivado IP integrator, the memory interface appears as discrete ports.

- `bram` : Implements array arguments as a standard RAM interface. If you use the RTL design in the IP integrator, the memory interface appears as a single port.
- `axis` : Implements all ports as an AXI4-Stream interface.
- `s_axilite` : Implements all ports as an AXI4-Lite interface. The HLS tool produces an associated set of C driver files during the Export RTL process.
- `m_axi` : Implements all ports as an AXI4 interface. You can use the `config_interface` command to specify either 32-bit (default) or 64-bit address ports and to control any address offset.
- `ap_ctrl_none` : No block-level I/O protocol.

Note: Using the `ap_ctrl_none` mode might prevent the design from being verified using the C/RTL co-simulation feature.

- `ap_ctrl_hs` : Implements a set of block-level control ports to start the design operation and to indicate when the design is `idle` , `done` , and `ready` for new input data.

Note: The `ap_ctrl_hs` mode is the default block-level I/O protocol.

- `ap_ctrl_chain` : Implements a set of block-level control ports to start the design operation, continue operation, and indicate when the design is `idle` , `done` , and `ready` for new input data.

Note: The `ap_ctrl_chain` interface mode is similar to `ap_ctrl_hs` but provides an additional input signal `ap_continue` to apply back pressure. Xilinx recommends using the `ap_ctrl_chain` block-level I/O protocol when chaining the HLS tool blocks together.

- `port=<name>` : Specifies the name of the function argument, function return, or global variable which the `INTERFACE` pragma applies to.

TIP: Block-level I/O protocols (`ap_ctrl_none` , `ap_ctrl_hs` , or `ap_ctrl_chain`) can be assigned to a port for the function return value.

- `bundle=<string>` : Groups function arguments into AXI interface ports. By default, the HLS tool groups all function arguments specified as an AXI4-Lite (`s_axilite`) interface into a single AXI4-Lite port. Similarly, all function arguments specified as an AXI4 (`m_axi`) interface are grouped into a single AXI4 port. This option explicitly groups all interface ports with the same `bundle=<string>` into the same AXI interface port and names the RTL port the value specified by `<string>`.

IMPORTANT: When specifying the `bundle=` name you should use all lower-case characters.